

Carina Zheng

AI Solution Architect | Machine Learning | Cloud & AI Systems

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PROFESSIONAL SUMMARY

AI Solution Architect with 8+ years of experience designing and delivering scalable AI/ML systems across cloud, edge, and enterprise environments. Proven ability to translate business requirements into end-to-end solutions, including data pipelines, model deployment, and production architectures.

Strong background in computer vision, data engineering, and MLOps, with expertise in building high-performance, cost-efficient AI systems.

CORE SKILLS

AI Solution Architecture

- End-to-end AI/ML system design (cloud, edge, hybrid)
- Scalable data pipelines and ML workflow architecture
- Model deployment and productionization
- Performance optimization and cost-efficiency strategies

Cloud & Data Engineering

- AWS cloud architecture and distributed systems
- ETL pipelines and large-scale data processing
- API design and system integration
- Data modeling and workflow orchestration

MLOps & DevOps

- CI/CD pipelines (Jenkins, Concourse, Docker)
- Model versioning and reproducibility (Git, DVC, MLFlow)
- Automated testing and deployment pipelines
- Production monitoring and reliability

Leadership & Stakeholder Engagement

- Stakeholder communication and requirement translation
- Cross-functional leadership in Agile/SCRUM environments
- Technical strategy and solution design leadership
- Business alignment and value-driven delivery

PROFESSIONAL EXPERIENCE

Intermediate Data Scientist – [Plexure](#) (Oct 2023 – Present)

- Designed and delivered scalable AI/ML solutions aligned with business objectives using Databricks
- Built end-to-end data pipelines and ML architectures across environments
- Collaborated with stakeholders to deploy AI systems

AI Engineer / Data Scientist – [EROAD](#) (Jul 2022 – Oct 2023)

AI Engineering

- Architected and delivered an end-to-end Driver Monitoring System (DMS) spanning cloud and edge, enabling real-time driver behaviour detection and privacy-preserving face anonymisation
- Designed scalable computer vision inference architectures, optimized for low-latency and resource-constrained edge environments
- Defined and implemented distributed data pipelines for various types data, supporting large-scale model training and real-time inference
- Built and managed production-grade AWS infrastructure, including automated CI/CD pipelines, improving deployment velocity, reliability, and system observability
- Led ML system design and optimisation, balancing performance, cost, and scalability across cloud and edge deployments
- Established engineering standards, code quality frameworks, and review processes across the department to improve delivery consistency and maintainability
- Collaborated with cross-functional stakeholders to translate business requirements into scalable AI solutions and technical roadmaps
- Founded and led an internal AI community of practice, driving knowledge sharing, innovation, and AI adoption across the organization

Senior Data Scientist: Modelling, Prediction and Data Visualization

- Develop prediction model to predict
- Build Data Visualization reports by using PowerBI, Shiny

Machine Learning Engineer – [Cubic Defence New Zealand](#) (Nov 2020 – Jul 2022)

- Develop a real-time object detection application on Jetson NX in a project team environment
- Design and implement project data capturing task on the field by using RGB and thermal cameras
- Plan, research and execute cutting-edge domains' solutions (e.g in algorithms, optimization, GPU utilization, pipeline) to advance client's needs
- Use Python, Torch and NVIDIA developer products to train, evaluate, deploy and maintain customized models
- Build and deploy project Docker images to meet development requirements
- Prototype hardware and software solutions for modeling and detection
- Research and prototype techniques and algorithms for object detection on Jetson and Diamond systems
- Contribute to preparing various product demonstrations for clients

Data Scientist – [INTELLIGENZ SOLUTIONS](#) (Nov 2019 – Feb 2020)

- Build a classification model to predict users' behaviours
- Design the project architecture
- Reports prediction result to clients via PowerBI reports
- Coordinate with technical lead to finish entire project architectural setup

Data System Analyst, Data Custodian – [Statistics New Zealand](#) (2016 – 2020)

- Designed large-scale data pipelines and ETL workflows
- Develop and manage data pipelines for large dataset
- Efficiently discover and derive hidden information from the datasets
- Perform data wrangling tasks on existing and new data
- Cooperating with commercial banks to provide financial provision to banks, government departments and the general public by using R and AWS
- Work in a fast-paced and agile environment that is responsive to customer needs
- Project-based and Organization-wide duties
- Lead a small team to deliver project designs, implementations and technical development
- Led technical delivery and stakeholder engagement

Database Analyst and Support Technician – [INTELLIGENZ SOLUTIONS](#) (Nov 2011 - Oct 2016)

- Develop and enhance modules in the software in C#
- Debug code errors and implement hotfixes
- Train and support domestic and international clients on-site
- Manage clients' and company's databases and ensure they have high data integrity
- Integrate data from various database types into the software's required database structure
- Deliver business and technical support to customers

EDUCATION

- **Master of Applied Data Science, Distinction** (GPA 7.3/9) – University of Canterbury
- **Bachelor of Science (Statistics & Computer Science)** – University of Canterbury

Summer Scholarship Project 2015/16 with Statistics New Zealand: A statistically informed interactive application on public commute

- **Diploma in Information and Communications Technology, CPIT**